

## Annex 1 - RDKit Full Descriptor List

Descriptors used in Machine Learning (ML) model development (Table S1 and S2) retrieved from RDKit software. The respective bibliographical foundation of each described is described in detail in the RDKit WebBook Documentation<sup>1</sup>. The descriptor calculations were made by converting each SMILES string (representing each molecular entry), running on top of Python version 3.9.8<sup>2</sup> and using RDKit package, version 2025\_03\_4 (Q1 2025)<sup>3</sup>. The Chemical information Type is presented, and the chemical species are sorted in the database file in the repository. The code used to generate descriptor values are in the online repository. The database uses as set of 106 for encoding values for each compound. The descriptors encode electronic, structural and van der Waals surface area (VSA) features.

**Table S1**

List of RDKit Descriptors used in ML model development ( $n = 106$  descriptors).

| Descriptor                      | Chemical information Type | Meaning                                                                                                                                                                                 |
|---------------------------------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BalabanJ (1)                    | Structural                | Distance sum of the two end-vertex for each edge. Connectivity of a molecule based on its graph structure (BalabanJ index has been proven to be relevant to network branching).         |
| BertzCT (1)                     | Structural                | Complexity index, considering both the variety of kinds of bond connectivity's and atom types; information contents related to bond connectivity and atom type diversity.               |
| $0\chi, 1\chi$ (2)              | Structural                | This descriptor signifies a retention index derived directly from gradient retention times. Considers individual atoms and their valence (0) and pairs of directly connected atoms (1). |
| $0\chi_n - 4\chi_n$ (5)         | Structural                | This descriptor signifies a retention index derived directly from gradient retention times (normalized indexes).                                                                        |
| $0\chi_v - 4\chi_v$ (5)         | Structural                | This descriptor signifies atomic valence connectivity index. Indices weighted by atomic valence                                                                                         |
| EState_VSA1 – EState_VSA11 (11) | VSA                       | MOE-type (QSAR model) descriptors using EState indices and surface area contributions. RDKit automatically calculates the EState index for each                                         |

|                                                      |            |                                                                                                                                                             |
|------------------------------------------------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                      |            | atom in the molecule, assigning each atom to a predefined EState range (bin). Sums the VSA of atoms within each bin (surface area grouped by EState range). |
| <b>Hall Kier <math>\alpha</math> (1)</b>             | Structural | Descriptor fingerprint that displays the difference between active and inactive molecules.                                                                  |
| <b>HeavyAtomCount (1)</b>                            | Structural | The number of heavy atoms in the molecule.                                                                                                                  |
| <b>HeavyAtomMolWt (1)</b>                            | Structural | The average molecular weight of the molecule ignoring hydrogens.                                                                                            |
| <b><math>\kappa_1, \kappa_2, \kappa_3</math> (3)</b> | Structural | This descriptor signifies # $\kappa$ shape index:<br>$(n-1) \times 2 / m^2$                                                                                 |
| <b><i>Max and Min AbsEStateIndex</i> (2)</b>         | Electronic | Maximum and minimum absolute E-State                                                                                                                        |
| <b><i>Max and Min AbsPartialCharge</i> (2)</b>       | Electronic | Maximum and minimum absolute partial charge                                                                                                                 |
| <b><i>Max and Min EStateIndex</i> (2)</b>            | Electronic | Maximum and minimum E-State                                                                                                                                 |
| <b><i>Max and Min PartialCharge</i> (2)</b>          | Electronic | Maximum and minimum partial charge                                                                                                                          |
| <b>Mol logP (1)</b>                                  | Structural | Wildman-Crippen logP value.                                                                                                                                 |
| <b>MolMR (1)</b>                                     | Electronic | Wildman-Crippen molar refractivity.                                                                                                                         |
| <b>MolWt (1)</b>                                     | Structural | The average molecular weight of the molecule.                                                                                                               |
| <b>NHOH Count (1)</b>                                | Structural | The number of NHs or OHs.                                                                                                                                   |
| <b>NO Count (1)</b>                                  | Structural | The number of Nitrogens and Oxygens.                                                                                                                        |
| <b><i>n<sub>alcarb</sub></i> (1)</b>                 | Structural | The number of aliphatic carbocycles.                                                                                                                        |
| <b><i>n<sub>alihet</sub></i> (1)</b>                 | Structural | The number of aliphatic heterocycles.                                                                                                                       |
| <b><i>n<sub>alirig</sub></i> (1)</b>                 | Structural | The number of aliphatic rings.                                                                                                                              |
| <b><i>n<sub>arocarb</sub></i> (1)</b>                | Structural | The number of aromatic carbocycles.                                                                                                                         |
| <b><i>n<sub>arohet</sub></i> (1)</b>                 | Structural | The number of aromatic heterocycles.                                                                                                                        |
| <b><i>n<sub>arorig</sub></i> (1)</b>                 | Structural | The number of aromatic rings.                                                                                                                               |

|                                        |            |                                                                                                                                                                                                                                        |
|----------------------------------------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b><math>n_{Ha}</math> (1)</b>         | Structural | The number of Hydrogen Bond Acceptors.                                                                                                                                                                                                 |
| <b><math>n_{Hd}</math> (1)</b>         | Structural | The number of Hydrogen Bond Donors.                                                                                                                                                                                                    |
| <b><math>n_{het}</math> (1)</b>        | Structural | The number of Heteroatoms.                                                                                                                                                                                                             |
| <b><math>n_{radele}</math> (1)</b>     | Structural | The number of radical electrons.                                                                                                                                                                                                       |
| <b><math>n_{rot}</math> (1)</b>        | Structural | The number of rotatable bonds.                                                                                                                                                                                                         |
| <b><math>n_{satcarb}</math> (1)</b>    | Structural | The number of saturated carbocycles.                                                                                                                                                                                                   |
| <b><math>n_{sathet}</math> (1)</b>     | Structural | The number of saturated heterocycles.                                                                                                                                                                                                  |
| <b><math>n_{satrig}</math> (1)</b>     | Structural | The number of saturated rings.                                                                                                                                                                                                         |
| <b><math>n_{ele}</math> (1)</b>        | Structural | The number of valence electrons.                                                                                                                                                                                                       |
| <b>PEOE_VSA1 – PEOE_VSA14 (14)</b>     | VSA        | MOE-type (QSAR model) descriptors using partial charges estimated using the Gasteiger PEOE and surface area contributions (sum the VSA of atoms within a given partial charge range).                                                  |
| <b>SMR_VSA1 – SMR_VSA10 (10)</b>       | VSA        | MOE-type (QSAR model) descriptors using atomic contributions to molar refractivity and surface area contributions (sums the VSA of atoms in each bin).                                                                                 |
| <b>SlogP_VSA1 – SlogP_VSA12 (12)</b>   | VSA        | MOE-type (QSAR model) descriptors using logP (calculated using the Crippen method) contributions and surface area sum of surface areas of atoms whose logP contribution falls within a specific range).                                |
| <b>TPSA (1)</b>                        | VSA        | The total polar surface area of a molecule based upon fragment calculations.                                                                                                                                                           |
| <b>VSA_EState1 – VSA_EState10 (10)</b> | VSA        | MOE-type (QSAR model) descriptors using EState indices and surface area contributions. The sum of VSA contributions of atoms whose EState values fall into specific predefined ranges (surface area grouped by predefined VSA ranges). |

## Annex 2 – Hyperparameter Section of Optimized Random Forest Model

*Random Forest Regressor* = (n\_estimators = 300, min\_samples\_leaf = 1 , max\_features = None, max\_depth = None, random\_state = 47).

## Bibliography

- (1) *RDKit: Open-source cheminformatics - Descriptor Guide Online Webbook*. <https://www.rdkit.org/docs/GettingStartedInPython.html#list-of-available-descriptors> (accessed July 1<sup>st</sup>, 2025).
- (2) *Python Software Foundation - Python Language Reference, version 3.9.8*. <http://www.python.org> (accessed July 1<sup>st</sup>, 2025).
- (3) Landrum, G. *RDKit: Open-source cheminformatics 2025\_03\_4 (Q1 2025) Release - June 2025*. <http://www.rdkit.org/> (accessed July 1<sup>st</sup>, 2025).